VEDH SONAWANE

IT Programmer

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SUMMARY

- Passionate about exploring technology, coding, and robotics
- Proficient in Python, Javascript, Bash, SQL, HTML, CSS, and Scratch
- Skilled in breaking down coding concepts for young learners of all ages
- Expertise in Google Suite (Docs, Slides, Sheets), Microsoft Office Suite (Word, Excel, PowerPoint)
- Experienced in leveraging AI models and integrating them into real-world projects
- Fluency in English, French, and Hindi

TECHNICAL SKILLS

- Programming Languages : Scratch, JavaScript, Python, SQL, HTML, and CSS
- RDBMS : Postgres SQL, SSMS
- Version Control : GitHub
- Game Development : Unity, Python (console games), Block coding (Scratch, Code.org, Blockly, etc.)
- API : Pull, Push, JSON, REST, HTTP requests
- Digital Communication : Zoom, Google Meet, Microsoft Teams
- Artificial Intelligence (AI) : Large Language Models (LLMs), OpenAI API, Prompt Engineering, Computer Vision (OpenCV), Neural Network Logic, TensorFlow, and Predictive Analytics.

EXPERIENCE

Application Developer / Lead Sensei @ Code Ninjas
Present

- Developed Python-based mini-projects (automation scripts, data exercises, etc.) to reinforce core programming concepts for students.
- Mentored 100+ students on Python libraries such as Pygame and Pandas, expanding their exposure to real-world applications.
- Applied toolsets consisting of JavaScript, Microsoft MakeCode Arcade, C#, and Unity, contributing to a 20% growth in student enrollments.
- Delivered engaging programming bootcamps and lessons for students, focusing on interactive learning and project-based experiences.
- Guided learners through troubleshooting and debugging logic errors, root cause analysis (RCA), improving problem-solving skills, code reliability, and scalability.

PROJECTS

- **Neural-Flux: AI Stress Simulator | Unity, C#, TensorFlow Created at DeltaHacks**
  - Developed an immersive 3D simulation in Unity designed to visualize the computational "pressure" and resource bottlenecks AI models face during real-time data processing.
  - Engineered a dynamic environment where "neural nodes" react to fluctuating data loads, using C# to script complex particle systems that represent data latency and hardware thermal throttling.

- Implemented a visual dashboard within the game to track "Model Burnout," providing a gamified educational experience on the importance of efficient code optimization in AI deployment.

- Bio-Sync: Real-Time Posture Correction via Computer Vision | Python, OpenCV, MediaPipe Created at Hack the North

- Built a health-tech application that utilizes a webcam to monitor a developer’s spinal alignment and ergonomic health in real-time.
- Leveraged the MediaPipe library to track 33 skeletal landmarks, calculating the angular deviation of the neck and shoulders against a "healthy" baseline.
- Designed an automated alert system that triggers subtle desktop notifications when poor posture is detected for more than 30 seconds, aimed at preventing long-term RSI (Repetitive Strain Injury) for remote workers.

- Eco-Ledger: Blockchain-Verified Carbon Credit Tracker | JavaScript, Solidity, React Created at DeerHacks

- Built a web application that helps local community gardens track and visualize their carbon reduction efforts.
- Used Firebase to store and manage project data securely, creating a transparent, easy-to-access record of environmental contributions.
- Designed a React-based front-end that allows users to log activities, see cumulative carbon impact, and share achievements with the community.
- Implemented simple verification features, like image uploads and admin approval, to ensure reported data is accurate and credible.
- Goal: Encourage local sustainability initiatives and raise awareness about carbon reduction in an accessible, engaging way, connecting communities through technology.

SOFT SKILLS

- Communication : Able to communicate effectively in person and online
- Teamwork : Dependable collaborator who can be counted on
- Time management : Skilled in goal setting, task prioritization, and time management to meet deadlines
- Decision-making : Proficient in analyzing information to support constructive decision-making and execution solutions
- Process-oriented : Committed to safe and efficient equipment handling by following guidelines and procedures
- Positive : Creates a positive learning environment and motivates learners to stay engaged

VOLUNTEERING EXPERIENCE

- Founder, Python Club: Designed and led Python lessons for students of all ages, teaching coding from the ground up in school community
- Learning Buddies: Mentored and engaged younger students through activities, while supporting positive behavior and supervision
- Assisted classmates with coding challenges during class, providing guidance and support in school club sessions
- Member, Leadership Club: Organized school events for holidays, ensuring engaging and age-appropriate activities for children aged 5–12
- Wee Walkers Club Volunteer: Guided children safely to their classrooms while monitoring their well-being
- Pizza Helpers Volunteer: Delivered orders to children in school, ensuring hygiene, safety, and proper handling
- Provided constructive feedback on classmates work to support learning and improvement
- Organized and shelved books and library materials in community centres, maintaining a clean and orderly space
- Assisted in organizing birthday parties and special events in the local community

AWARDS & CERTIFICATIONS

- PCEP: Python Certified Entry-Level Programmer
- Awarded Mathalon Knowledgehook Medal for expertise in Math - 2023
- Founder, Python Club Recognition (School Recognition)

REFERENCES

Provided upon request